

M. Mudaffarsyah

Software Developer & Machine Learning Engineer

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SUMMARY

Machine Learning Engineer and Software Developer with hands-on experience in developing AI-powered web, mobile, and backend applications. Skilled in building end-to-end machine learning pipelines using PyTorch, developing scalable backend services with FastAPI and Laravel, and creating modern web applications with Next.js. Experienced in integrating machine learning models into software applications, designing RESTful APIs, and working with SQL and NoSQL databases. Passionate about building intelligent, scalable, and impactful software solutions.

EDUCATION

Informatics Engineering, Raja Ali Haji Maritime University | Tanjungpinang, Indonesia

Bachelor of Engineering (Sep 2022 - Sep 2026)

- **GPA** (3.7/4.0)
- **Relevant Courses:** Machine Learning, Web Programming, Mobile Application Development, Object-Oriented Programming, Design and Analysis of Algorithms, Distributed Systems.
- **Thesis Title:** Performance Analysis of Local Training in Federated Learning for Waste Classification Using ResNet-18

PROFESSIONAL EXPERIENCE

Dinas Komunikasi dan Informatika Kota Tanjungpinang | Tanjungpinang, Indonesia

Web Developer (Jul 2025 - Sep 2025)

- Collaborated with a two-member team to design and develop government web applications.
- Redesigned the UI/UX and developed the frontend for the Website Kelurahan Tanjungpinang Kota.
- Designed and developed the Website Survei Kepuasan Masyarakat (SKM) from UI/UX design to backend implementation, which is currently used by the Tanjungpinang City Government.

Raja Ali Haji Maritime University | Tanjungpinang, Indonesia

Research Assistant (Jul 2024 - Nov 2024)

- Preprocessed image datasets for machine learning model development and evaluation in De Grove, an AI-powered mobile application for mangrove species identification.
- Contributed to frontend and backend development of the De Grove mobile application within a multidisciplinary research team.
- Supported the development of an ecotourism application used to identify mangrove species in Desa Pengudang.

Practice Assistant (Mar 2024 – Jun 2024)

- Facilitated database laboratory sessions by teaching database concepts, guiding SQL implementation, and supporting students through hands-on practical exercises.
- Mentored students throughout the practicum by providing technical guidance, answering questions, facilitating discussions, and supervising competency assessments.

RESEARCH

Performance Analysis of Federated Learning System (Thesis) | Tanjungpinang, Indonesia

Software Developer & ML Engineer (Sep 2025 - Jun 2026)

- Enhanced the Waste Sort Android application by redesigning the user interface, refining the application workflow, and implementing features to support Federated Learning research.

- Integrated PyTorch-based machine learning models with FastAPI backend services and developed an administration dashboard for managing Federated Learning workflows, including training configuration and global model management.

PROJECTS & SKILLS

- **Projects:** Waste Sort: Federated Learning-Based Waste Classification Application (2026), DeGrove: AI-Powered Mangrove Species Identification Mobile Application (2024), Predict: Sales and Inventory Forecasting System Using Linear Regression (2024), Website Survei Kepuasan Masyarakat (SKM) (2025), QIQO: Real-Time Interactive Quiz Application (2024).
- **Technical Skills:**
 - **Web & Backend Development:** Laravel, Next.js, Vue.js, FastAPI, PHP, JavaScript, RESTful API
 - **Mobile Development:** Kotlin, Jetpack Compose, Flutter
 - **Machine Learning & AI :** Python, PyTorch, TensorFlow, Deep Learning, Computer Vision.
 - **Database & Tools :** Git, GitHub, MySQL, MongoDB, MVC, MVVM
- **Languages:** Intermediate in English, Native in Indonesian.